

Rahul Krishna K

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EDUCATION

Program	Institution	CGPA/%	Completion
BTech & MTech in Chemical Engineering	IIT Madras	8.29	2024
Semester Exchange in Computing Science	University of Groningen	7.70	2022
XII (CBSE Board)	KVN Public School	96%	2018

RESEARCH EXPERIENCE

- **Polymer Research Group, IIT Madras - Undergraduate Researcher** (June 2023 - May 2024)
 - Developed fully machine driven Polymer Informatics Pipeline inspired by NLP for Master's project
 - Trained a BERT based embedding model on polymer sequences to create molecular fingerprints
 - Achieved an accuracy of 0.80 on property prediction task by training on 2 million sequences compared to polyBERT(State-of-the-art model, June 2023) with accuracy of 0.81 trained on 100 million sequences.
- **TCS Research - Research Intern** (June 2022 - July 2022)
 - Predicted ingredient functionality mapping for small organic molecules from using Deep Learning
 - Achieved an accuracy of 0.76 on multi-label classification using SMILES2VEC, a CNN-LSTM model

PUBLICATIONS

- **Deep Learning Framework for Materials Property Prediction and Design Using QR Codes** (Nov, 2023)
 - Developed a QR code based molecular finger print, compared one hot encoding, Morgan finger print and QR-code based finger prints for property prediction. Visualized fingerprints after dimensionality reduction using UMAP. Co-authored a research paper, published in ACS Engineering AU.

WORK EXPERIENCE

- **Flexday.ai -Full time AI Engineer** (Aug 2024 - present)
 - Integrated tree-organized summary retrieval into RAG system enabling it to answer holistic questions
 - Implemented Late Integrations from BERT to improve retrieval and re-ranking latency by 20 times
 - Lead a team to develop self-service analytics by building text2SQL and text2Insight solutions
- **IIT Madras - Teaching Assistant** (Aug 2023 - Dec 2023)
 - Taught tutorial classes for the Master's Level Course 'Mathematical Foundations for Data Science'

PROJECTS

- **Placement Chatbot - Pet Project** (Sept, 2023)
 - Developed a data augmented chat bot to chat with placement and internship bluebooks
 - Assembled Llama 2 for text generation, Elastic Search for vector store and Streamlit for UI
- **Smart grading - AI freelancing project** (July, 2024)
 - Engineered a service for evaluating handwritten student answers using Foundation Models
 - Trained custom layout detection model (FasterRCNN) to identify student answers from answer sheets

COURSE WORK

- **Math:** Maths Foundations for Data Science , Probability & Statistics , Calculus for Artificial Intelligence
- **Data Science:** Intro to Machine Learning , Artificial Intelligence , Multivariate Data Analysis
- **Programming:** Design and Analysis of Algorithms , Programming & Data Structures in Python

PLATFORMS & PACKAGES

- **Data Science :** Pytorch, Scikit-learn, TensorFlow, Haystack, NLTK , Git, Docker, Amazon Web Service

CO-CURRICULAR AND EXTRA-CURRICULAR ACTIVITIES

- Active member of **Institute football team**, and NSO (National Sports Organization) Football at IIT Madras
- Motivated aspiring IITians through an inspirational talk at SEED Global Public school as Chief Guest